WEEK 2 INTERNSHIP REPORT

Cybersecurity Internship | DG Interns Hub

Student Name	_____
College / Institution	GITAM, Visakhapatnam
Programme / Track	Computer Science & Engineering – Cyber Security
Reporting Period	Week 2 (October 2026)
Submission Date	30 October 2026

Report overview

This report summarizes the cybersecurity learning activities, virtual lab exercises, network analysis, Windows event-log investigation, and supporting documentation completed during Week 2. It records observed results and identifies items that were not fully verified so that the submission remains accurate.

1. Objectives

The main objectives for the week were to understand common network protocols and scanning techniques, capture and interpret network traffic, perform controlled Nmap scans against a personally managed virtual machine, investigate Windows security events, and document findings in a structured SOC-style workflow.

2. Theory and learning notes

Study notes were prepared on TCP connection establishment and teardown, common network protocols, Wireshark capture and display filters, Nmap scan types, Windows fundamentals, Windows Security Event IDs, logon types, and the concept of establishing a system baseline.

3. TryHackMe practical learning

Writeups were completed for the following learning rooms:

- Further Nmap
- Live Host Discovery
- Basic Port Scans
- Windows Basics
- Windows Fundamentals 1
- Windows Fundamentals 2

The writeups document the key concepts and practical learning from each room and are intended to be stored separately in the Week 2 Writeups folder.

4. Network lab: Apache, connectivity and packet analysis

An Apache web server was installed and started on the Ubuntu virtual machine. The Apache service was confirmed as active. Kali Linux communicated with Ubuntu over the VirtualBox host-only network.

Observed configuration and results

Item	Observed result
Ubuntu host-only IP	192.168.56.103
Kali host-only IP	192.168.56.104
ICMP connectivity	Ping from Kali to Ubuntu succeeded (4 packets received, 0% packet loss).
HTTP connectivity	curl to the Ubuntu Apache server returned the default Apache page; HTTP response was 200 OK.
Packet capture	Wireshark on Kali captured TCP handshake traffic, HTTP traffic, ICMP traffic, and Nmap SYN packets.

Wireshark was used to inspect the traffic generated during connectivity and web-server tests. DNS traffic was not consistently observed in the available capture and should be marked as not verified unless a separate DNS capture is saved.

5. Nmap scanning exercise

Nmap was run from Kali Linux against the user's own Ubuntu virtual machine (192.168.56.103). The following scan types were performed and results were reviewed:

- Host discovery using `-sn` — target host reported as up.
- SYN scan using `-sS` — TCP port 80 reported open (HTTP); an additional closed port was reported.
- Service/version detection using `-sV` — Apache httpd 2.4.66 (Ubuntu) identified on TCP port 80.

• OS detection using `-O` — scan completed; Nmap did not identify an exact OS match.

An aggressive scan (`-A`) and a scan against a separate Windows virtual machine were not confirmed as completed. Only report these as completed if corresponding output files are present.

6. Windows event-log investigation

Windows Event Viewer and PowerShell were used to examine Security log activity on the Windows host. The investigation included reviewing successful and failed logon events and checking selected event IDs relevant to account use and privileged activity.

Events observed in the additional event-ID query

Event ID	Meaning	SOC relevance
4648	A logon was attempted using explicit credentials.	Can help identify use of supplied credentials by a user or process.
4672	Special privileges assigned to a new logon.	Useful for reviewing privileged logons and administrative activity.

Multiple occurrences of Event IDs 4648 and 4672 were observed around 29 October 2026 at approximately 16:58. These events are not, by themselves, proof of malicious activity; their account, process, logon, and privilege details should be correlated before drawing conclusions.

A PowerShell command was prepared to export selected Security event details to a text file. A separate export of Event ID 4625 failed-logon records to CSV was also created or verified at the Desktop path. The final copy of each export into the Findings folder should be checked before submission.

Note: the Security log was accidentally cleared during the exercise. The report should not claim that a complete historical event set was preserved. Any missing events or evidence should be described as unavailable rather than reconstructed.

7. Cheat sheets and documentation

The following reference materials were prepared:

• Wireshark Filter Cheat Sheet — 20 useful display filters with meanings and uses.
• Windows Event ID Cheat Sheet — 15 important security event IDs, meanings, and SOC investigation uses.
• Theory notes and individual TryHackMe room writeups.

8. Skills practised

• Linux service management and basic web-server validation.
• VirtualBox host-only networking and connectivity troubleshooting.
• Packet capture and protocol inspection with Wireshark.
• Host discovery, port scanning, service detection, and OS detection with Nmap.
• Windows Security log review and basic PowerShell event retrieval/export.
• Technical note-taking, evidence organization, and SOC-oriented interpretation.

9. Evidence checklist

Evidence	Status to verify before submission
Apache service and HTTP response screenshots	Confirm saved in Findings.

Evidence	Status to verify before submission
Wireshark PCAPNG capture(s)	Confirm capture files are saved in Captures; DNS evidence is not verified.
Nmap .txt and .xml outputs	Confirm all completed scan outputs are copied into Findings.
Windows Event Viewer screenshots	Include only screenshots actually captured; do not imply missing events exist.
4625 CSV and event findings TXT	Confirm both exports are copied into Findings.
Notes, Writeups and Cheat Sheets	Confirm all files are in their respective folders.

10. Conclusion

Week 2 provided practical exposure to network monitoring, basic reconnaissance, Windows event investigation, and technical documentation. The work connected foundational networking concepts with hands-on SOC-style activities. The main outcomes were successful Kali-to-Ubuntu connectivity, Apache HTTP validation, packet inspection, Nmap scan execution, and identification of Windows privileged-logon and explicit-credential events. Before final submission, all evidence files should be checked against the folder checklist and incomplete activities should remain clearly labelled.

Suggested folder structure

DG Interns Hub / Week 2 / Notes
DG Interns Hub / Week 2 / Writeups
DG Interns Hub / Week 2 / Findings
DG Interns Hub / Week 2 / Captures
DG Interns Hub / Week 2 / Cheat Sheets

Prepared as a draft from the activities and results reported during the conversation. Verify filenames, screenshots, and exported evidence before submission.